Project
Company Bankruptcy Prediction

Team Members:

- Bhushan Asati (Cluster 0)
- Anirudh Sharma (Cluster 1)
- Rujuta Dabke (Cluster 5)
- Suyash Madhavi (Cluster 6)

1. Introduction
In this project, we applied a "Divide and Conquer" strategy to predict company bankruptcy in a highly imbalanced dataset. Our approach involved:

- **Clustering:** Reducing features to around top 40 and using K-Means (K=7) to create homogeneous subgroups.
- **Stacking:** Building specialized Stacking Classifiers (Random Forest, K-Nearest Neighbors, Logistic Regression) for each subgroup containing bankruptcies.
- **Generalization:** Combining these expert models to predict bankruptcy on the test set, achieving a constraint of **4.15%**.

2. Company Characterization (Rubric Section 3.2)
We identified 7 distinct subgroups within the training data. Below is the summary of their distributions and characteristics.

Subgroup Statistics:

- **Cluster 0 (Standard):** 2,846 companies | 63 Bankrupt (2.21%). This is the largest group representing "average" financial health.
- **Cluster 1 (Safe):** 1,693 companies | 3 Bankrupt (0.18%). This group showed very stable financial metrics with minimal risk.
- **Cluster 5 (High Risk):** 727 companies | 129 Bankrupt (17.74%). This group contained the majority of bankruptcies.
    - *Characteristic:* Companies here consistently scored low on "Net Income to Total Assets" and "Persistent EPS".
- **Cluster 6 (Edge Case):** 3 companies | 3 Bankrupt (100%). A small outlier group of distressed companies.
- **Clusters 2, 3, 4 (No Risk):** Small groups (Total: 538 companies) with 0 historical bankruptcies. Modeled with a Constant Classifier to reduce False Positives.

3. Subgroup Bankruptcy Prediction (Rubric Section 3.3.2)

A. Results Table

Below is the performance table for our Stacking Models. The "True True" (TT) indicates correctly identified bankrupt companies.

Table 3: Model Performance by Subgroup

Subgroup ID	**Student Name**	**Num Companies**	**Num Bankrupt**	**Stacking TT (Correct)**	**Stacking TF (Missed)**	**N_{features}**
0	Bhushan	2,846	63	61	2	45
1	Anirudh	1,693	3	0	3	36
2	Constant	1	0	0	0	0
3	Constant	1	0	0	0	0
4	Constant	536	0	0	0	0
5	Rujuta	727	129	124	5	35
6	Suyash	3	3	3	0	40
Team Total	**--**	**5,807**	**198**	**188**	**10**	**41**

(Note: Team Total N_{features} is the weighted average).

B. Meta-Model vs. Base Models

Per the rubric, we compare the final Stacking Model (Meta) against its individual Base Models.

Cluster 0

- **Meta-Model**:
  Accuracy = 82.53%
- **Base Models**:
  RF Accuracy = 0% | GB Accuracy = 0.63% | LR Accuracy = 70.10%

Cluster 5

- **Meta-Model**:
  Accuracy = 96.1%
- **Base Models**:`
  RF Accuracy = 17.83% | GB Accuracy = 25.58% | LR Accuracy = 70.54%

Cluster 1

- **Meta-Model**: Predicted 0/3 Correctly (Due to extreme imbalance).
  Accuracy = 0%
- **Base Models**: All base models also predicted predominantly "Non-Bankrupt".
  RF Accuracy = 0% | LR Accuracy = 0%

Cluster 6

- **Meta-Model:** Constant Classifier predicted 3/3 Correctly (100% Accuracy).

4. Generalization Results
On the unseen Test Data, our combined model predicted **42 companies** as bankrupt out of 1012.

- **Positive Prediction Rate:** 4.15%
- **Constraint:** Passed (< 20%)

5. Video Presentation
This video demonstrates our code workflow, explains the "Divide and Conquer" logic, and showcases our final results.